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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I175*

Team Name: *Code Storm*

Domain: *Digital Technologies,AI and Cybersecurity*

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Problem Statement

When a critical system fails, every second costs time, money, and safety. Teams often discover equipment failures only after they happen, while disconnected IoT devices and growing cyber threats make systems harder to monitor and protect. What if a Digital Twin could mirror the real world in real time, AI could predict failures before they happen, IoT could continuously sense every change, and Cybersecurity could detect threats before they cause damage?

Problem Understanding

- Who experiences this problem? :Plant operators, maintenance engineers, and cybersecurity teams working with critical systems like industrial machines, power plants, and healthcare equipment
- What causes the problem? :Disconnected monitoring systems, unreliable sensor data, and cybersecurity vulnerabilities, so failures and cyberattacks are often detected only after damage has occurred.
- How does it affect users or stakeholders? :Unexpected failures cause financial losses, operational delays, and safety risks, and late detection makes early intervention difficult.

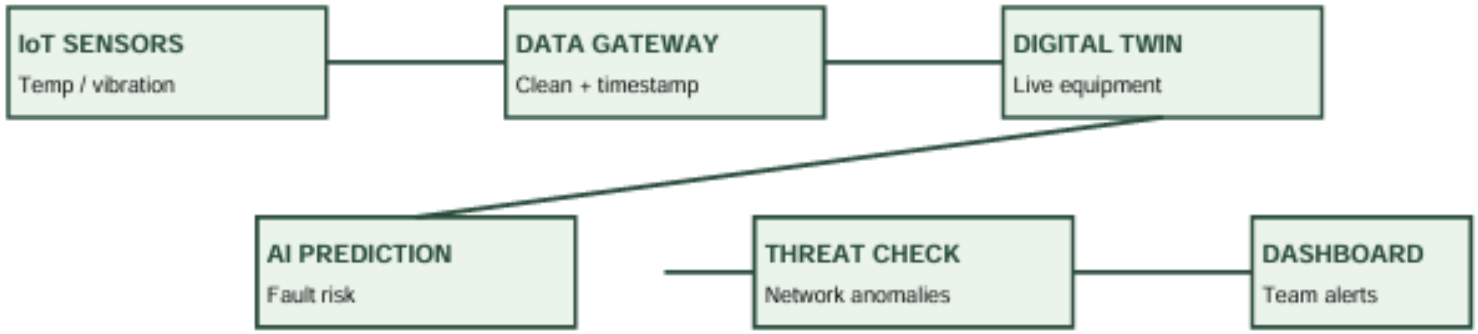
Existing Gaps & Challenges

- Existing Gap 1: Monitoring, maintenance, and cybersecurity tools work separately, while noisy or incomplete sensor data can reduce AI prediction accuracy.
- Existing Gap 2: Digital Twin and AI systems can be costly, and delays, limited explainability, and integration difficulties make real-time failure and cyber-threat detection challenging
- Main Challenge That Remains Unsolved: Developing an affordable, integrated, and scalable system that combines real-time equipment monitoring, accurate failure prediction, and cyber-threat detection with clear, actionable alerts.

Our Proposed Solution

- Core Idea or Product An integrated smart monitoring system that combines IoT sensors, Digital Twin technology, AI-based predictive maintenance, and cybersecurity in one platform.
- How the Solution Works at a High Level IoT sensors collect equipment data continuously. AI analyzes the data to predict possible failures, while the Digital Twin mirrors equipment conditions and cybersecurity monitoring detects suspicious network activity. A unified dashboard displays warnings and equipment status.
- Main Value Delivered to Users Early warnings, improved maintenance decisions, faster cyber-threat detection, reduced downtime and costs, and better safety and reliability of critical systems.

Technical Approach



Sensors feed the gateway, which updates the Digital Twin. AI estimates equipment fault risk, while the security layer checks for unusual activity. The dashboard combines both alerts for human review.

Tech Stack

Frontend: HTML/CSS/JavaScript dashboard

Backend: Python (FastAPI)

Database: SQLite / PostgreSQL

AI/ML, APIs, or Hardware: scikit-learn / TensorFlow / ESP32 + sensors or simulated data

Innovation & Uniqueness

- What is new or different? :TwinGuard combines Digital Twin technology, AI-based predictive maintenance, IoT monitoring, and cybersecurity in one integrated platform instead of using separate tools.
- Why is this approach suited to the problem?: It addresses disconnected monitoring, unexpected equipment failures, unreliable sensor data, and cyber threats by combining real-time monitoring, failure prediction, and threat detection.
- What is the key differentiator?: Its key differentiator is a unified dashboard that provides both equipment-health predictions and cyber-threat alerts, with an emphasis on affordable, scalable deployment.

Feasibility & Scalability

- **Feasibility:** Use affordable IoT sensors, Python AI models, Digital Twin simulation, a database, and a web dashboard. Build and test each module, then integrate them to predict failures and detect cyber threats
- **Scalability:** Start with one machine and expand to multiple machines or industrial facilities by adding sensors, AI models, and cybersecurity features.
- **Risks:**High costs can be reduced with open-source tools; inaccurate predictions with better data and testing; cyber threats with encryption and access controls; processing delays with edge computing; and integration issues with standard communication protocols.

Cost for Product (Optional)

- Development or hardware cost: 10,000–25,000 for IoT sensors, a microcontroller, and prototype development.
- Hosting or maintenance cost: Cost: 0–1,000 per month using open-source tools and free-tier hosting.
- Estimated total: - 10,000–25,000 for the initial prototype.

Use Cases

- Use case 1:Plant operator in a manufacturing unit. A motor's vibration and temperature start rising slowly. TwinGuard's AI predicts a likely failure and the dashboard warns the operator early, so the machine can be repaired before it breaks down.
- Use case 2:Maintenance engineer at a power plant. The engineer checks the Digital Twin dashboard to see the live condition of turbines and generators. They schedule maintenance only where the fault risk is high, which saves time and cost.
- Use case 3: Cybersecurity team at a hospital. Unusual network activity is detected on connected medical equipment such as ventilators. The team gets an alert and blocks the threat before it affects patient safety.

Impacts and Benefits

- Benefit to Target Users: Helps plant operators, maintenance engineers, and cybersecurity teams detect failures and cyber threats early, make better decisions, and reduce equipment downtime
- Social, Economic, or Environmental Impact: Improves workplace safety, reduces financial losses and maintenance expenses, and minimizes energy and material waste caused by equipment failures
- Expected Measurable Outcome: Aim to reduce unexpected downtime, improve fault and threat detection speed, lower maintenance costs, and provide earlier, actionable alerts. Actual improvements will be measured during prototype testing.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	TwinGuard makes critical infrastructure more reliable and resilient. It predicts equipment failures and detects cyber threats early, using affordable IoT and AI tools.
SDG 8	Target 8.8	Decent Work and Economic growth	Early failure and threat alerts make workplaces safer for plant operators and maintenance

Individual Member Contribution

Member	Contribution or responsibility
Yakavi S	Problem research and gaps
Aathmika S R	Core idea, solution, AI and security approach, and PPT
Akshayashree ST	Tech stack and system flow
Kamireddy Harshini	Cost, feasibility, risks, and impact

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